

## Adrenal Hormones: Response to Stress

The adrenal glands of vertebrates play a major role in the response to *stress*, a state of threatened homeostasis. Located atop the kidneys (the *renal* organs), each **adrenal gland** is actually made up of two glands with different cell types, functions, and embryonic origins: the adrenal *cortex*, the outer portion, and the adrenal *medulla*, the central portion (**Figure 45.19**). The adrenal cortex consists of true endocrine cells, whereas the secretory cells of the adrenal medulla develop from neural tissue. Thus, like the pituitary gland, each adrenal gland is a fused endocrine and neuroendocrine gland.

### The Role of the Adrenal Medulla

Imagine that while walking in the woods at night you hear a growling noise nearby. “A bear?” you wonder. Your heart beats faster, your breathing quickens, your muscles tense, and your thoughts speed up. These and other rapid responses to perceived danger comprise the “fight-or-flight” response. This coordinated set of physiological changes is triggered by two hormones of the adrenal medulla, epinephrine (adrenaline)

and **norepinephrine** (also known as noradrenaline). Both are *catecholamines*, a class of amine hormones synthesized from the amino acid tyrosine. Both molecules also function as neurotransmitters, as you’ll read in Concept 48.4.

As hormones, epinephrine and norepinephrine increase the amount of chemical energy available for immediate use (see Figure 45.19a). Both catecholamines increase the rate of glycogen breakdown in the liver and skeletal muscles and promote the release of glucose by liver cells and of fatty acids from fat cells. The released glucose and fatty acids circulate in the blood and can be used by body cells as fuel.

Catecholamines also exert profound effects on the cardiovascular and respiratory systems. For example, they increase heart rate and stroke volume and dilate the bronchioles in the lungs, actions that raise the rate of oxygen delivery to body cells. For this reason, doctors may prescribe epinephrine as a heart stimulant or to open the airways during an asthma attack. Catecholamines also alter blood flow, causing constriction of some blood vessels and dilation of others. The overall effect is to shunt blood away from the skin, digestive organs, and kidneys while increasing the blood supply to the heart, brain, and skeletal muscles.

▼ **Figure 45.19** Stress and the adrenal gland.

